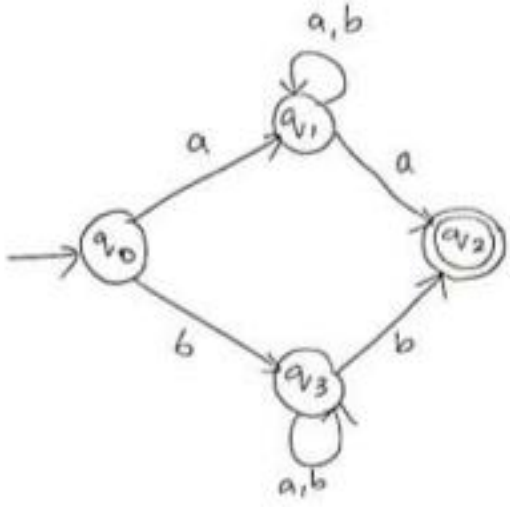
# SAVEETHA SCHOOL OF ENGINEERING

# SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

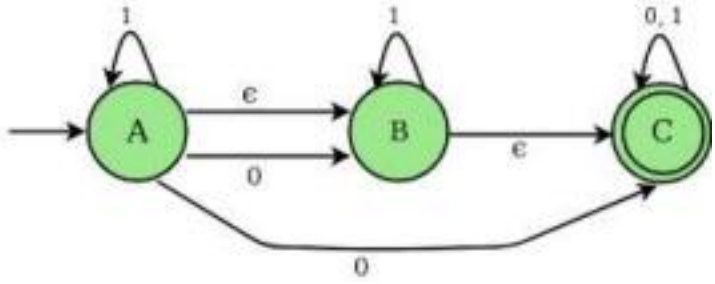
## COMPUTER SCIENCE AND ENGINEERING

## CSA13 – THEORY OF COMPUTATION

## ASSIGNMENT – 1

| | | CO |
|---|---|---|
| 1 | Design a Deterministic Finite Automata (DFA) to accept strings that start and end with same symbols over ΣΣ={0,1}. Write the formal definition of the DFA. Also show that the string **010100** is accepted by the DFA | 1 |
| 2 | Design NFA to accept string having 101 as substring | 1 |
| 3 | Apply the principle of mathematical induction and prove that for all positive integers n, the statement given below is true. $$\sum_{nn=1}^{nn} nn(nn+1) = nn(nn+1)(nn+2)3$$ | 1 |
| 4 | Consider the NFA given below which accepts the language representing the set of all strings that start and end with the same symbol over the alphabet ΣΣ={a,b}. Construct a DFA that represents the same language   | 1 |

| | | |
|---|---|---|
| 5 | A NFA with ε-transitions is given below. Find ε-closure for each state and construct an equivalent NFA without ε-transitions that accepts the same language. Draw its transition table and mark the initial and final states | 1 |

| | | |
|---|---|---|
| |  | |

**1. Design a DFA to accept strings that start and end with the same symbols over Σ = 1.**

Given:

Alphabet: Σ = 1

Language: All strings that start and end with the same symbol.

Example: 010100

**AIM**

To design a Deterministic Finite Automaton (DFA) that accepts strings starting and ending with the same symbol over Σ = 1.

**1. Formal Definition**

A DFA is defined as a 5-tuple:

M = (Q, Σ, δ, q0, F)

Where:

Q = Σ = 1

q0 = Initial state

F = (Final states)

δ = Transition function.

**2. Transition Table**

| Present State | Input 0 | Input 1 |
|---|---|---|
| → q0 | q1 | q2 |
| * q1 | q1 | q3 |
| * q2 | q3 | q2 |
| q3 | q1 | q2 |

Note:

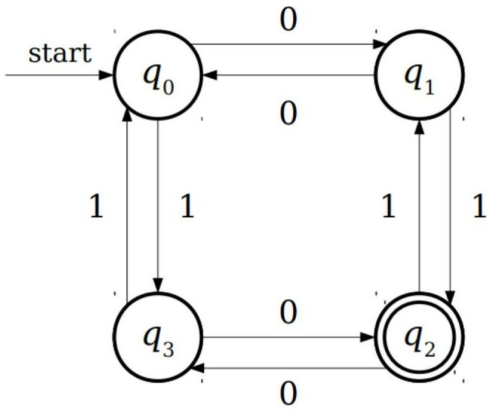
q0 = Initial state

q1 = Starts with 0 and ends with 0 (Final state).

q2 = Starts with 1 and ends with 1 (Final state).

q3 = Starts and ends with different symbols.

**3. DFA Diagram**




**4. Verification of String 010100**

Given string:

W = 010100

Step-by-step transitions:

| Step | Input | State |
|---|---|---|
| 1 | Start | q0 |
| 2 | 0 | q1 |
| 3 | 1 | q3 |
| 4 | 0 | q1 |

| | | |
|---|---|---|
| 5 | 1 | q3 |
| 6 | 0 | q1 |
| 7 | 0 | q1 |

Final state = q1.

Since q1 is a final state:

**ACCEPTED**

$010100 \in L(M)$

**RESULT**

Thus, the DFA accepting strings that start and end with the same symbols is designed successfully. The given string 010100 is accepted because it starts with 0 and ends with 0.

**2. Design an NFA to accept strings having 101 as a substring over Σ = 1.**

**AIM**

To design a Non-Deterministic Finite Automaton (NFA) that accepts strings containing 101 as a substring.

**1. Formal Definition**

An NFA is defined as a 5-tuple:

$M = (Q, \Sigma, \delta, q0, F)$

Where:

$Q = \Sigma = 1$

$q0$ = Initial state

F = (Final state)

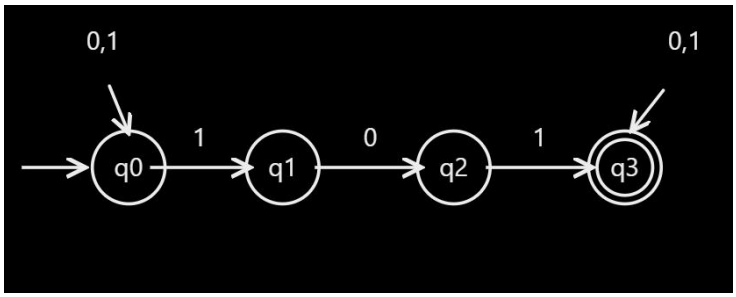
$\delta$ = Transition function.

**2. Transition Table**

| **Present State** | **Input 0** | **Input 1** |
|---|---|---|
| → q0 | | |
| q1 | | ∅ |
| q2 | ∅ | |

* q3

**3. NFA Diagram**




q0 has a loop on 0 and 1; q3 has a loop on 0 and 1. q3 is the final state.

**4. Explanation**

q0 reads any starting symbols (0 or 1).

When it reads 1, it can move to q1.

q1 reads 0 and moves to q2.

q2 reads 1 and moves to q3.

q3 accepts any remaining symbols (0 or 1).

**5. Verification**

Given string:

W = 1101010

It contains the substring 101.

One accepting path:

1 0 1 0 1 0

q0 ---> q1 ---> q2 ---> q3 ---> q3 ---> q3 ---> q3

Final state = q3.

**STRING ACCEPTED**

**RESULT**

Thus, the NFA for accepting strings containing 101 as a substring is designed successfully. The given string 1101010 is accepted.

**3. Apply the Principle of Mathematical Induction and prove the given statement for all positive integers n.**

Given:

sum_{r=1}^{n} r(r+1) = n(n+1)(n+2) / 3

**AIM**

To prove the given mathematical statement using the Principle of Mathematical Induction (PMI).

**Proof**

Step 1: Basis Step (n = 1)

LHS:

sum_{r=1}^{1}r(r+1) = 1(1+1)=2

RHS:

1(1+1)(1+2) / 3 = (1)(2)(3) / 3 = 2

Therefore,

**LHS = RHS**

Hence, true for n = 1.

Step 2: Induction Hypothesis

Assume the statement is true for n = k.

sum_{r=1}^{k}r(r+1) = k(k+1)(k+2) / 3

Step 3: Induction Step

We need to prove the statement for n = k + 1.

sum_{r=1} ^ {k+1}r(r+1)
= sum_{r=1} ^ {k} r(r+1) + (k+1)(k+2)

Using the induction hypothesis,

= (k(k+1)(k+2) / 3) + (k+1)(k+2)

Taking common terms,

= ((k+1)(k+2) / 3)[k+3]
= (k+1)(k+2)(k+3) / 3

This is the required expression for n = k + 1.

Step 4: Conclusion

Since the statement is true for n = 1 and true for n = k + 1 whenever it is true for n = k, by the Principle of Mathematical Induction, the statement is true for all positive integers n.

sum_{r=1}^{n}r(r+1) = n(n+1)(n+2) / 3

PROVED

**4. Convert the given NFA into an equivalent DFA.**

Given:

Alphabet: Σ =

Initial state: q0

Final state: q2

NFA states: Q =

**AIM**

To construct an equivalent Deterministic Finite Automaton (DFA) for the given NFA using the subset construction method.

**Step 1: NFA Transition Table**

| State | a | b |
|---|---|---|
| → q0 | | |
| q1 | | |
| q3 | | |
| * q2 | ∅ | ∅ |

**Step 2: Find DFA States (Subset Construction)**

Let the new DFA states be:

A =

B =

C =

D =

E =

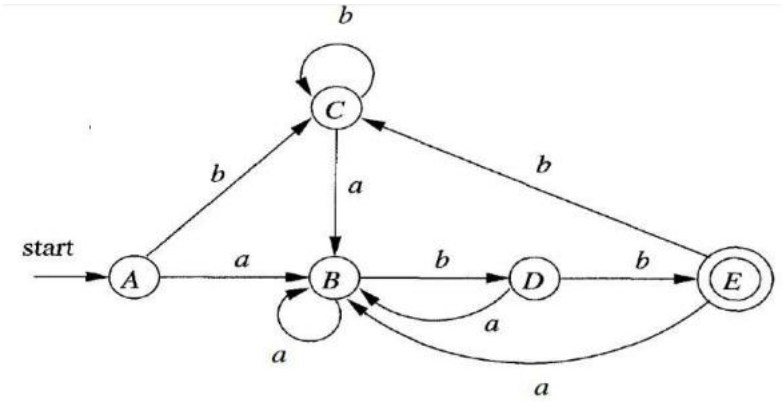
Initial state = A

Final states = D, E (because they contain q2).

**Step 3: DFA Transition Table**

| DFA State | a | b |
|---|---|---|
| → A | B | C |
| B | D | B |
| C | C | E |
| * D | D | B |
| * E | C | E |

**Step 4: Equivalent DFA Diagram**




For drawing in your record, use these transitions:

A --a→ B

A --b→ C

B --a→ D

B --b→ B (self-loop)

C --a→ C (self-loop)

C --b→ E

D --a→ D (self-loop)

D --b→ B

E --a→ C

E --b→ E (self-loop)

Final states: D and E (draw double circles).

**Step 5: Result**

Thus, the equivalent DFA is constructed successfully using the subset construction method.

Initial state: A

Final states: D, E

Total DFA states: 5

**5. Remove ε-transitions from the given NFA and construct an equivalent NFA.**

**AIM**

To find the ε-closure of each state and construct an equivalent NFA without ε-transitions.

**Step 1: Given NFA**

States: Q =

Alphabet: Σ = 1

Initial state: A

Final state: C

**Step 2: Find ε-closure of each state**

1. ε-closure(A)

From A, we can move to B using ε and then B to C using ε.

ε - closure (A) = {A,B,C}

2. ε-closure(B)

From B, we can move to C using ε.

ε - closure (B) = {B,C}

3. ε-closure(C)

There is no ε-transition from C.

ε - closure © = {C}

**Step 3: Construct NFA without ε-transitions**

Formula:

δ'(q,a) = ε - closure (δ( ε - closure (q),a))

Calculations:

For A:

δ’(A,0) = {B,C}
δ’(A,1) = {A,B,C}

For B:

δ’(B,0) = {C}
δ’(B,1) = {B,C}

For C:

δ’(C,0) = {C}
δ’(C,1) = {C}

**Step 4: Transition Table**

| State | Input 0 | Input 1 |
|---|---|---|
| → * A | | |
| * B | | |
| * C | | |

Initial state: A

Final states: A, B, C

Note: All three states become final because their ε-closures contain the original final state C.

**Step 5: Equivalent NFA Diagram**

Draw the following transitions:

A → B, C on input 0

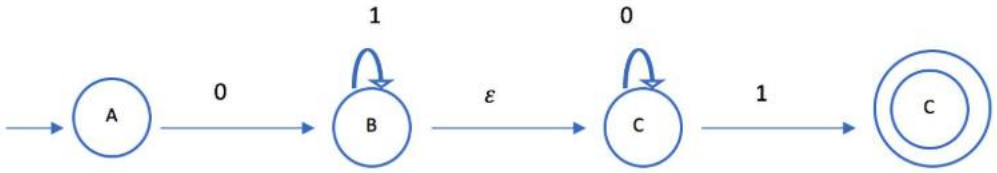
A → A, B, C on input 1

B → C on input 0

B → B, C on input 1

C → C on input 0 and 1 (self-loops)

ε – NFA:

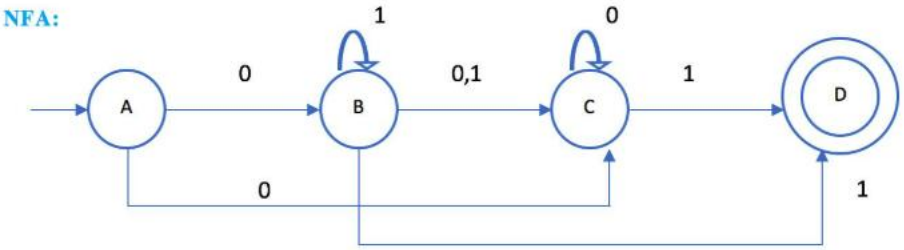



| | 0 | 1 |
|---|---|---|
| → A | {B, C} | ∅ |
| B | {C} | {B, C, D} |
| C | {C} | {D} |
| * D | ∅ | ∅ |

| | ε* | 0 | ε* |
|---|---|---|---|
| A | A | B | B C |
| B | B C | ∅ C | C |
| C | C | C | C |
| D | D | ∅ | ∅ |

| | ε* | 1 | ε* |
|---|---|---|---|
| A | A | ∅ | ∅ |
| B | B C | B D | B C D |
| C | C | D | D |
| D | D | ∅ | ∅ |

NFA:




**RESULT**

Thus, the ε-closure of all states is calculated and the equivalent NFA without ε-transitions is constructed successfully.

Initial state: A

Final states: A, B, C

ε-transitions: Removed.